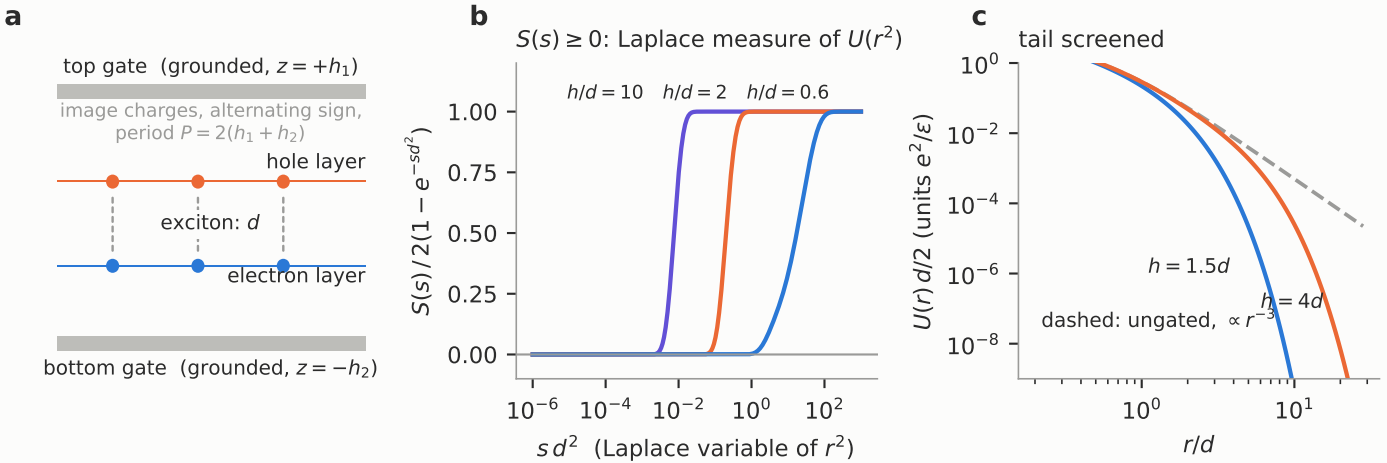
a
top gate (grounded, $z = +h_1$)
image charges, alternating sign,
period $P = 2(h_1 + h_2)$
hole layer
exciton: d
electron layer
bottom gate (grounded, $z = -h_2$)
b
$S(s) \geq 0$: Laplace measure of $U(r^2)$
$h/d = 10$ $h/d = 2$ $h/d = 0.6$
$S(s)\,/\,2(1 - e^{-sd^2})$
1.00
0.75
0.50
0.25
0.00
10^{-6} 10^{-4} 10^{-2} 10^{0} 10^{2}
$s\,d^2$ (Laplace variable of r^2)
c
tail screened
$U(r)\,d/2$ (units e^2/ε)
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
$h = 1.5d$
$h = 4d$
dashed: ungated, $\propto r^{-3}$
10^{0} 10^{1}
r/d